

FPGA Side-Channel Analysis

Final Presentation

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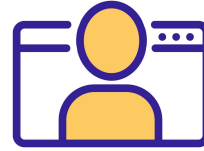
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Agenda

Project Summary

- Top Level Project Description/Purpose Statement
- Motivation

Methods and Procedures

- Methods, Procedures, and Plans
- Schedule
- Deliverables

Findings

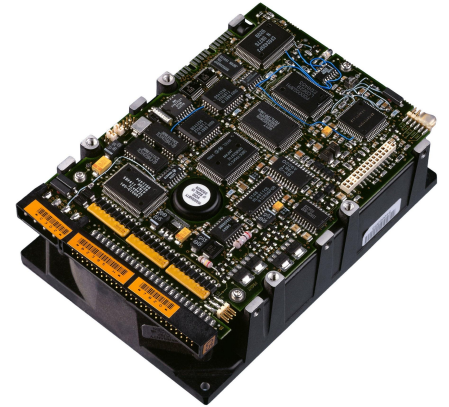
Issues and Challenges

Conclusions and Recommendations

Project Summary

Purpose Statement and Motivation

- FPGAs are fundamental hardware building blocks
 - Their security is crucial - supply chain attacks!
- Goal: Provide a reliable means to detect malicious modifications to loaded programs



Methods and Procedures

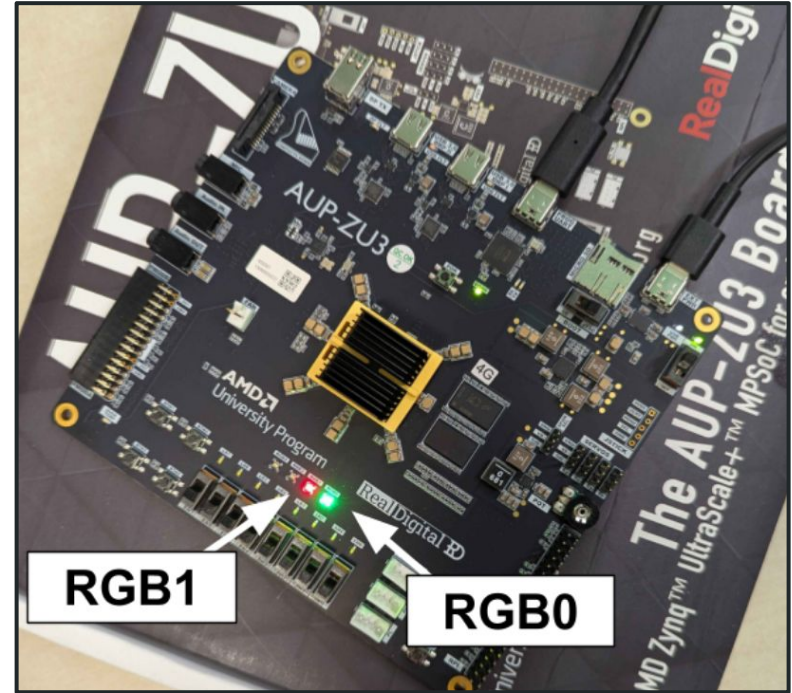
Methods and Procedures Used

- Development of two program pairs
 - Traffic Light and Elevator
 - Each pair: A benign version and a modified version
- Power Analysis
 - Power meter - FNIRSI FNB48S placed between outlet and board
 - Data collection - FNIRSI USB data logger on Ubuntu VM
 - <https://github.com/baryluk/fnirsi-usb-power-data-logger>
- EM Measurements
 - HackRF SDR and H-Loop Probe
 - Custom Python script to conduct wide- and narrow-spectrum sweeps



The Programs

- Program 1: Simple finite-state "traffic light" machine that performs the following:
 - RGB0, RGB1 both red for 1 second
 - RGB0 green for 10 seconds.
 - RGB0 yellow for 1 second
 - RGB0, RGB1 both red for 1 second
 - RGB1 green for 10 seconds.
 - RGB1 yellow for 1 second
 - (the cycle repeats.)



The FPGA we are using (RealDigital's AUP-ZU3 board), with the traffic light program running.

The Programs

- Program 2: Elevator Simulator
 - Pushbuttons to input desired floor
 - LEDs to indicate current floor
 - RGBs to indicate elevator behavior
 - RGB0: Going up
 - RGB1: Going down
 - RGB2: Door is open (idle)
- Program modification
 - Extraneous floating-point operations every clock cycle

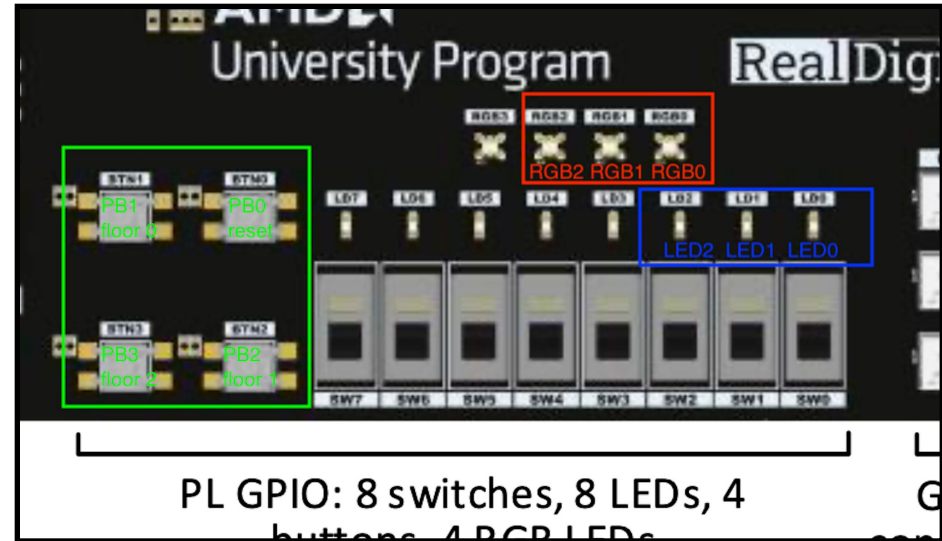


Diagram showing the key parts of the board for the elevator program.

Power Measurements

Trials

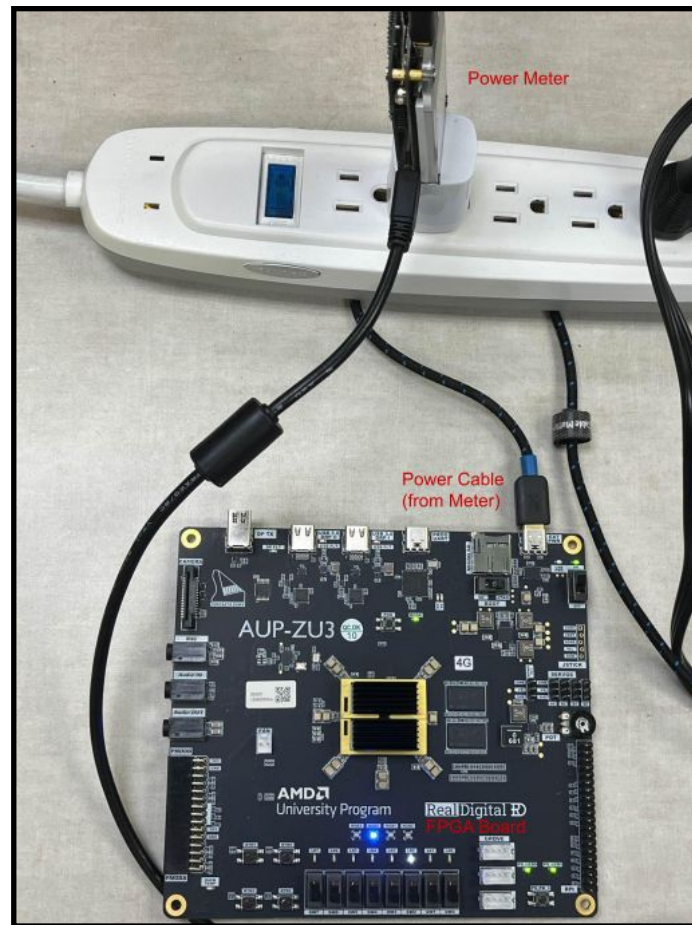
- 3 trials for each benign program
- 3 trials for each modified program
- 45 seconds per trial

Traffic Light

- Trials begin at the start of the first cycle

Elevator

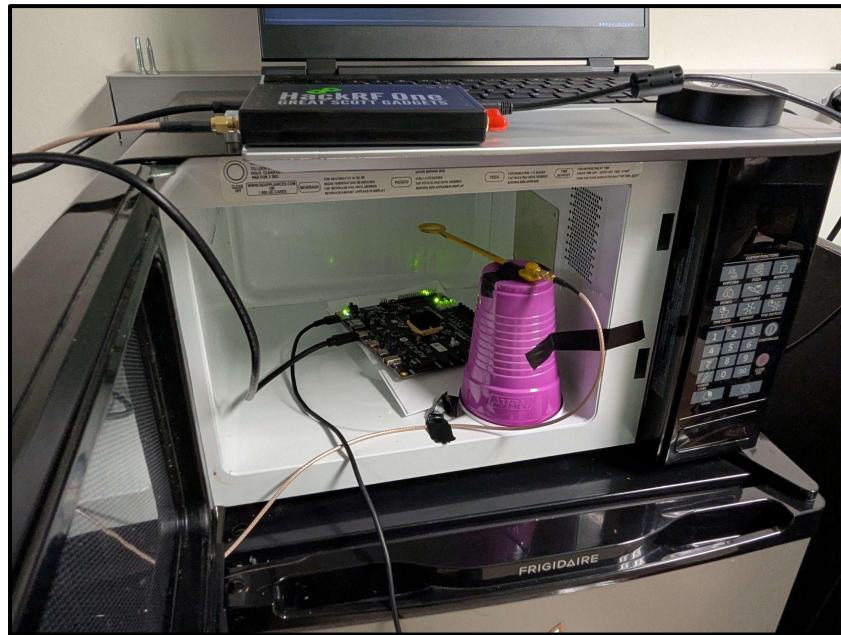
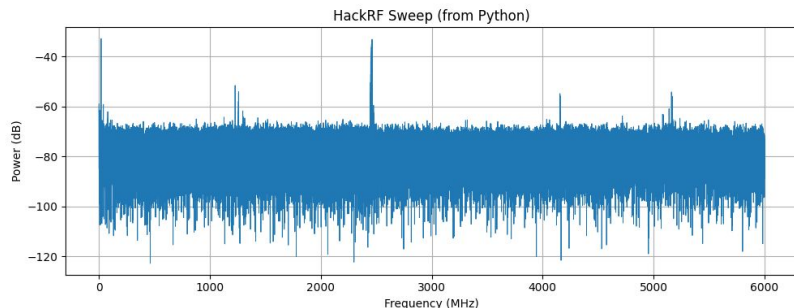
- Trials begin at default state (floor 0)
- Change to floor two 20 seconds into the recording



Labeled setup for power analysis, with the elevator program running.

Electromagnetic Side-Channel Analysis Workflow

- Capture EM emissions using HackRF and H-Loop Probe
- Sweep frequency bands for each program to locate harmonic anomalies
- Zoom in on identified anomalies
- Compare signatures between malicious and known-good workloads



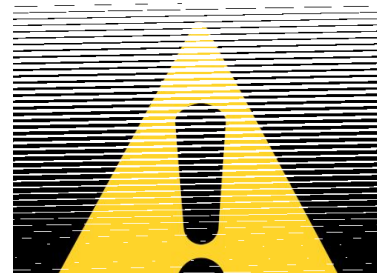
Our very advanced setup.

Schedule and Deliverables

Date	Task Completed
9/26	Initial draft of benign traffic light program completed
10/8	Reception of hardware (FPGAs, ChipWhisperers, HackRF)
10/17	Final completion and loading of benign traffic light program
10/29	Completion of benign elevator program and high-level synthesized version of benign traffic light program
10/31	Midterm report and presentation
11/7	Completion and loading of modified versions of the benign programs
11/14	Setup for side-channel measurements begins
11/21	Side-channel measurements for both the benign programs and modified programs taken
11/28	Drafting of final report and presentation
12/12	Side-channel measurement data
12/12	Final report and presentation

Limitations and Delimitations

- Unsophisticated, inexpensive setup
 - FNB48S not as precise as ChipWhisperer
 - Setup means data might not be entirely accurate
 - Hardware limitations: possible interference
- Noise and possible EM interference
- Scope: proof-of-concept
 - Yes/no question on whether SCA can detect trojan
 - Small sample size (3 runs x 4 programs = 12 samples)
 - All we need is a ballpark estimate



Findings

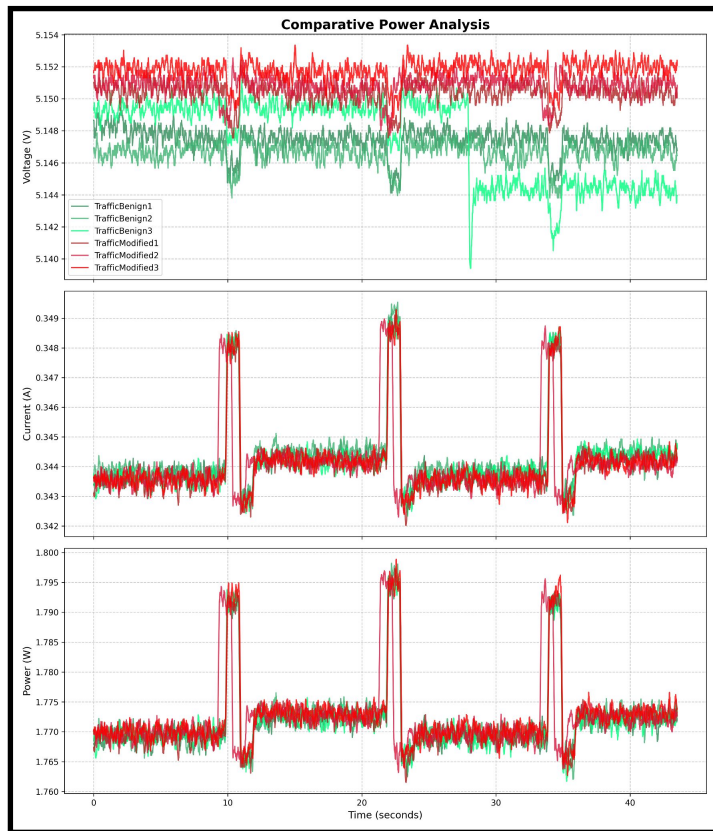
Power Measurements

Fairly consistent results

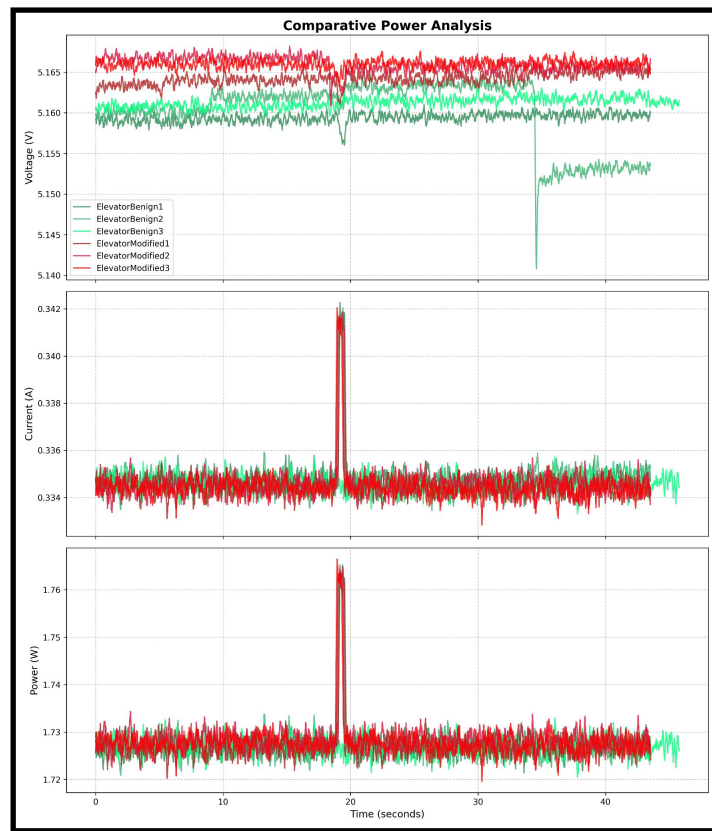
- Voltage: Clear and consistent differences
 - Modified program has higher voltage
- Current and Power: No significant variation

Program	Avg. ΔV	Avg. ΔA	Avg. ΔW
Traffic Light	+0.011V	-0.0005A	+0.0011W
Elevator	+0.015V	-0.0005A	+0.0025W

Power Measurements

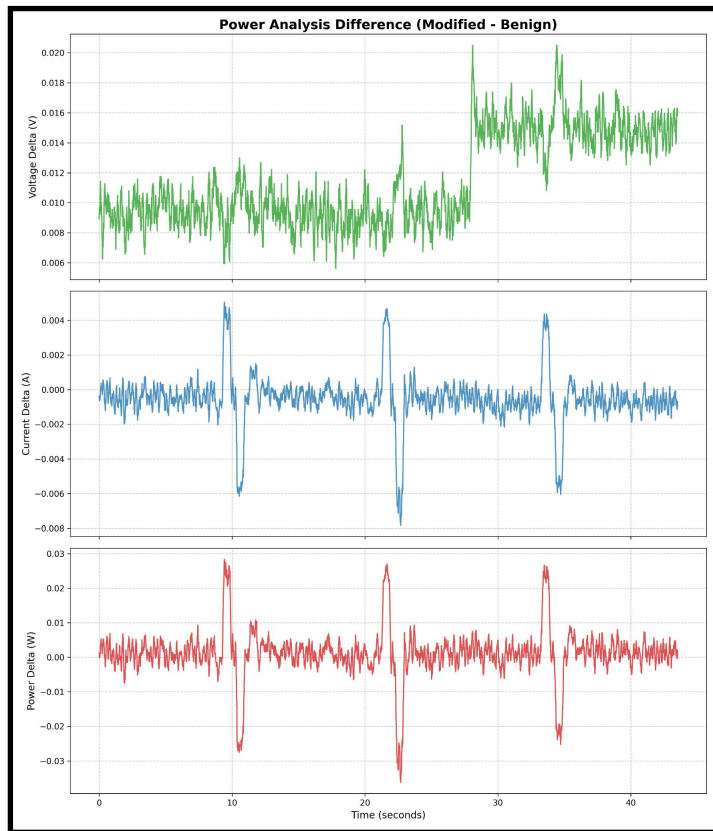


Power analysis measurements for the benign (green) and modified (red) traffic light programs.

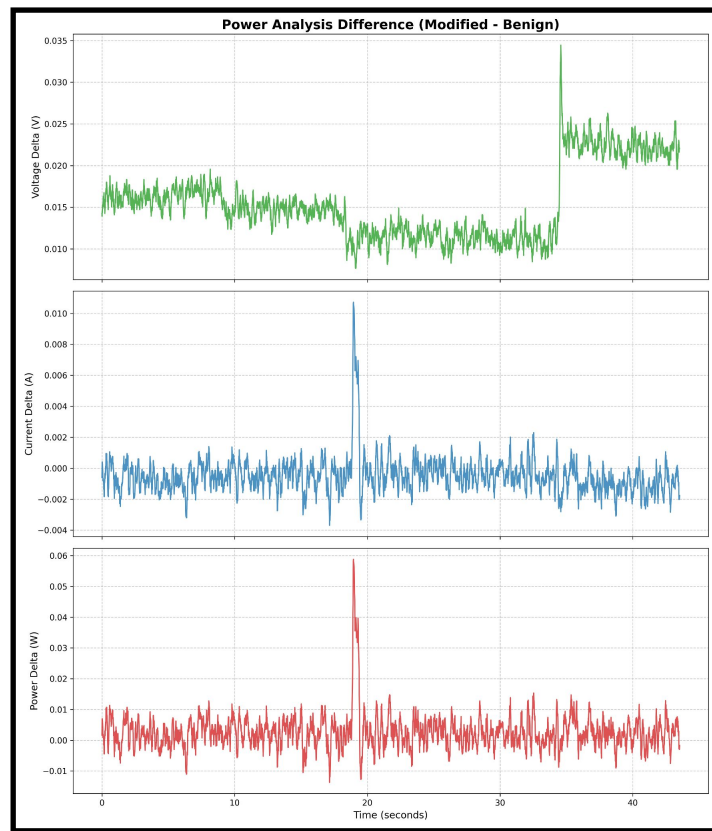


Power analysis measurements for the benign (green) and modified (red) elevator programs.

Power Measurements



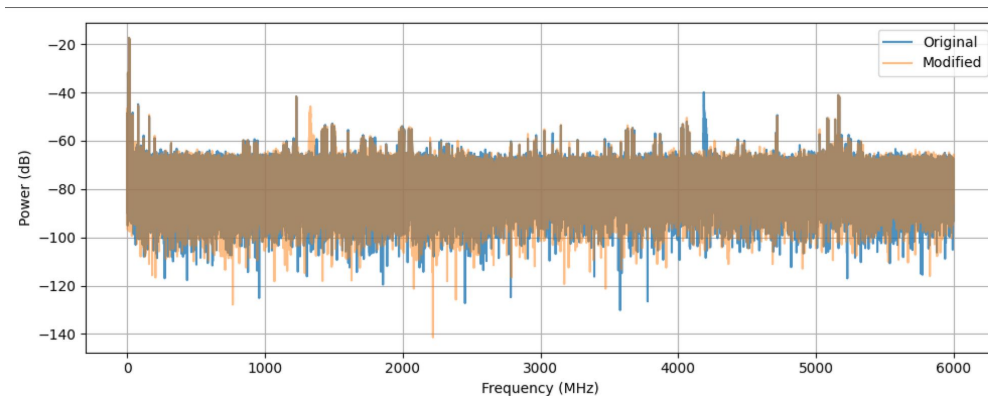
Power difference measurements for the traffic light programs.



Power difference measurements for the elevator programs.

Electromagnetic Side-Channel Analysis - Sweeps

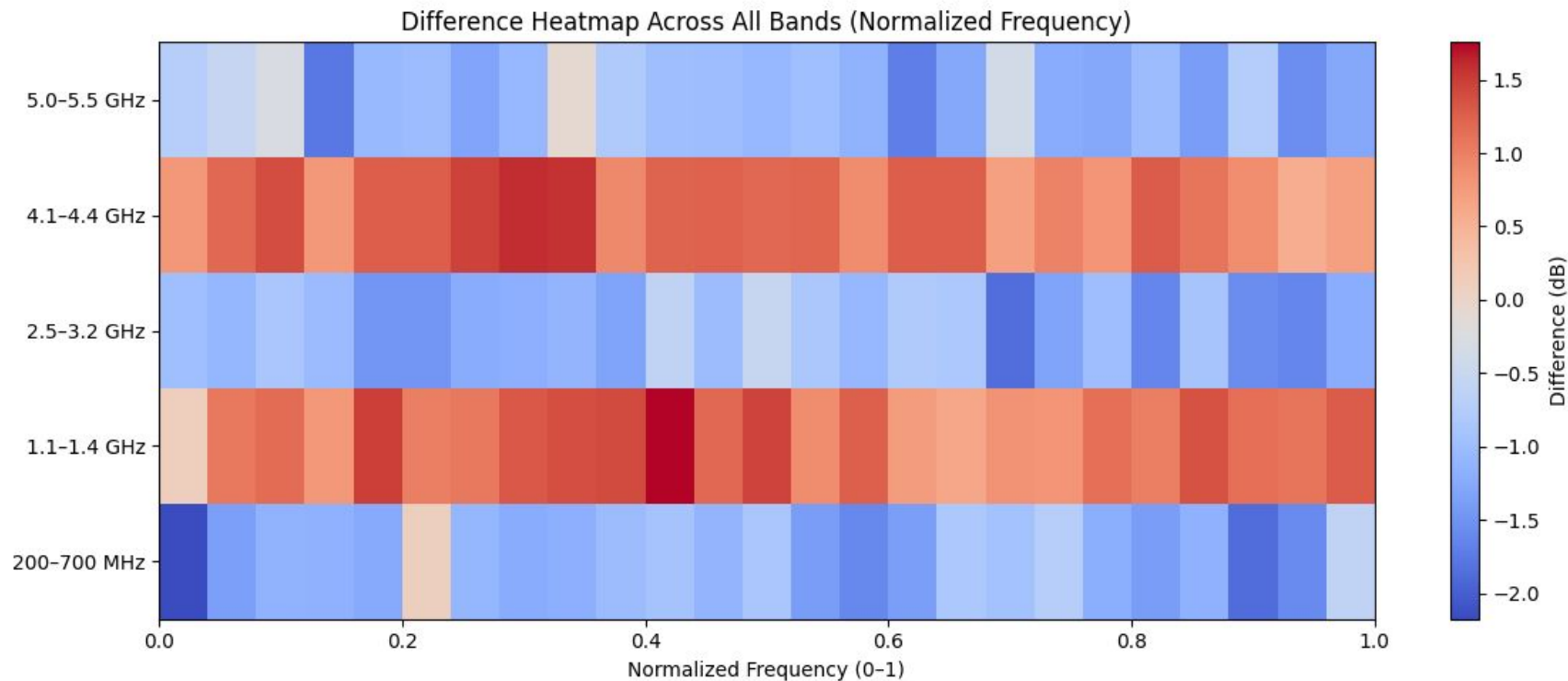
- Sweeps conducted between 1 to 6000 MHz
- Averaging 10 sweeps for each benign and modified program
- Notable differences around:
 - 200-700 MHz
 - 1.1-1.4 GHz
 - 2.5-3.2 GHz
 - 4.1-4.4 GHz
 - 5.0-5.5 GHz



Electromagnetic Side-Channel Analysis - Narrow Bins

- RF emissions collected across each of the notable frequency bands
- Each run was a random time delta apart
- The differences appear as localized peaks between 1.3-2.2 dB in the selected bands
 - Exceed the run-to-run noise level expected (i.e. not just noise)
- Differences occur in expected CPU harmonic regions
- Differences do not prove malicious behavior but do show measurable changes in RF emissions.

Electromagnetic Side-Channel Analysis - Narrow Bins



Normalized heatmap showing EM emission changes across bands

Issues, Challenges, and Conclusions

Issues and Challenges

- Hardware shipping delays
- Hardware troubleshooting
- Learning how to program FPGAs
- Failed initial power analysis setup
 - Planned to use ChipWhisperer for measurement
 - Soldering required but unsuccessful
 - Pivoted to FNIRSI FNB48S instead
- EM Analysis is still very manual



Conclusions and Recommendations



- Voltage analysis reliably detects modification
 - Not power or current
- Recommend taking repeated voltage measurements to detect difference
 - How many measurements?
 - What is “enough” of a difference?
- Electromagnetic emissions also reliably detect statistical differences across all expected harmonics
 - ML classifier to further analyze the emissions
 - Analysis with different parameters on the SDR
 - Scientific microwave and solo cup worked well, additional noise reduction recommended
- Future work - try different setups

Appendix

Bibliography

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7. Sayakkara, Asanka, et al. "Leveraging electromagnetic side-channel analysis for the investigation of IOT devices." Digital Investigation, vol. 29, July 2019, <https://doi.org/10.1016/j.diin.2019.04.012>.

Acronyms

- **EM** - Electromagnetic
- **EM-SCA** - Electromagnetic side-channel analysis
- **FPGA** - Field-programmable gate array
- **FSM** - Finite state machine
- **HLS** - High level synthesis
- **SCA** - Side-channel Analysis
- **SDR** - Software-defined radio

Key Terms

- **Hardware Trojan:** A malicious modification to a circuit or logic that is disguised as being a part of the circuit, but has some harmful function.
- **FPGA:** AKA Field-Programmable Gate Array - is a configurable integrated circuit that can be continuously reprogrammed by users to perform some function.
- **Bitstream:** A file containing the data to be loaded onto the FPGA - this includes the main FPGA logic (program) and configuration data for routing, memory, and I/O.
- **Side-Channel Analysis:** Obtaining (potentially sensitive) information from a system through measuring the side effects of its main computations, such as power consumption and EM emissions.
- **Signal Processing:** Extracting information from a signal (sound, images, sensor data)
- **Supply Chain:** The life cycle of a hardware component, where at each point an attacker can add some malicious function

Q&A